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Chapter 15
Player X Strikes Back: Full-Street Games

In the first part of Part III, we discussed half-street games. In these games, Player X acted first, but was compelled to check dark. After this Player Y could either check and take a showdown, or value bet and bluff appropriately. In Chapter 15, we begin our investigation of full-street games. In these games, Player X again acts first, but now can either bet or check. If he bets, Player Y will sometimes have the option to raise or fold, and can always call. If he checks, then Player Y can check or bet.

It is important to recognize that some concepts will carry over from half-street games, while others will not. For example, betting and bluffing ratios are just as valid for Player X as they were for Y. But in addition to this, the option that X has to bet ***changes the distribution of hands he has when he checks,*** so Y's play behind his checks will frequently change. This is a feature of virtually all expanded toy games - the play on earlier decisions significantly impacts the play on the later ones.

A second important idea is that ***Player X cannot lose value moving from a half-street game to a full-street one,*** assuming that the rest of the game is the same. If Player X's strategies are such that he has less expectation in the full-street game than the half-street one. he should simply adopt the strategy previously forced upon him of checking in the dark. This is a natural extension of the idea that strategic options have non-negative value - X's ability to bet right off the bat can at worst be worth zero, since he can always decline the option and take his half-street equity.

The third and most important idea that applies to full-street games, and to any game of more complexity than a half-street game, is the concept and the way in which a strategy is expressed. In half-street games, particularly the [0,1] game, each player had only one action. Player Y could bet or check, and if Player Y bet. Player X could call or fold. Since each of their strategies was fundamentally just a single action in the game, there was a direct relationship between the ***strategy*** "fold" and the ***strategic option*** "fold."

In the half street games, each player's strategy was made up of regions - sometimes these regions were a range of hands with which the player would use a particular strategic option. This occurred in games like the [0,1] game. In other cases, the regions were mixing frequencies for the discrete hands - such as in the AKQ game. But in either case, the player selected a distribution of hands with which to take each strategic option, and the total of those distributions comprised the strategy.

In full-street games, though, the situation is slightly different, hi half-street games. Player X's initial check was required by the rules of the game. But now he has what might be called two different decisions - to check or bet initially, and to call or fold to a bet if he is faced with that scenario. In our view, these are not really two separate decisions. Instead, when X chooses a strategy, we create a new set of more complex options that reflect his choices for the game. Instead of two options on the first decision and two on the second, X simply has three decisions - to bet, to check-call, or to check-fold. When we specify decisions for full-street games, we will create payoff matrices that reflect the expanded strategic options and not simply the immediately available ones.

With these ideas in mind, we introduce our first full-street game.

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Example 15.1 - AKQ Game #4

One full street.
Pot size of P units. (P > 1 for the same reason as in AKQ Game #1 - Example 13.1.)
Limit betting of 1 unit.
Raising is not allowed.

The complete ex-showdown payoff matrix for this game as follows:

	Y	**Ace**				**King**				**Queen**			
		X bets		X checks		X bets		X checks		X bets		X checks	
X		Call	Fold	Bet	Chk	Call	Fold	Bet	Chk	Call	Fold	Bet	Chk
Ace	Bet					-1	0			-1	0		
	Chk/Call							-1	0			-1	0
	Chk/Fold							+P	0			+P	0
King	Bet	+1	-P							-1	0		
	Chk/Call			+1	0							-1	0
	Chk/Fold			0	0							+P	0
Queen	Bet	+1	-P			+1	-P						
	Chk/Call			+1	0			+1	0				
	Chk/Fold			0	0			0	0				

We can again immediately remove dominated options from this game to create a simpler game. For Player X, we can see that checking and folding an ace is dominated by checking and calling with it, and checking and calling with a queen is dominated by checking and folding it. For Player Y, calling a bet with a queen or folding to a bet with an ace are dominated. Additionally, checking an ace behind a check is dominated by betting it. Removing these leaves the following game:

	Y	**Ace**		**King**				**Queen**		
X		Call	Bet	Call	Fold	Bet	Chk	Fold	Bet	Chk
Ace	Bet			-1	0			0		
	Chk/Call					-1	0		-1	0
King	Bet	+1						0		
	Chk/Call		+1						-1	0
	Chk/Fold		0						+P	0
Queen	Bet	+1		+1	-P					
	Chk/Fold		0			0	0			

In this simpler game, we can see that betting kings is dominated for both players, and so we have:

	Y	**Ace**	**King**		**Queen**	
X		Call/Bet	Call/Chk	Fold/Chk	Fold/Bet	Fold/Chk
Ace	Bet		-1	0	0	0
	Chk/Call		0	0	-1	0
King	Chk/Call	+1			-1	0
	Chk/Fold	0			-P	0
Queen	Bet	+1	+1	-P		
	Chk/Fold	0	0	0		

Let us consider the strategy choices for X. X will check all his kings, and call and fold with them to make Y indifferent to bluffing a queen. In addition to this, he must choose a strategy for his aces and for his queens. When he has an ace, he gains by check-calling when his opponent has a queen by ***including a bluff.*** However, he gains by betting when his opponent has a king and his opponent will check-call. With queens, he gains by bluffing out his opponent's kings and loses

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when his opponent holds an ace or a king that will call.

For Y, the picture is a little simpler. He has only to choose strategies for his kings and for his queens. With a king, he will call often enough to make X indifferent to bluffing with a queen. With queens, he will bluff often enough to make X indifferent to calling with a king.

The strategy for X, then, will consist of three values:
A betting frequency with aces, which we will call x.
A calling frequency with kings, which we will call c.
(Note that X checks with all his kings, so this is his check-calling frequency)
A bluffing frequency with queens, which we will call b.
The strategy for Y will consist of just two values:
A calling frequency with kings, which we will call c_y.
A bluffing frequency with queens, which we will call b_y.

Consider the situation when X bets. We know that X will only bet when he holds aces or queens. Y must make him indifferent to bluffing by calling with a particular frequency of kings, and this is true whether the first player bets all his aces or not. Just as in the half-street games, when a player bets a bifurcated set of aces and queens, his opponent must fold precisely a of his hands that beat bluffs. Using the same logic from AXQ Game #2 (Example 13.2), we can end that when X bets, Y must call with $(P- 1)/(P+ 1)$ of his kings so that X is indifferent to bluffing queens. This ensures that Y will be folding α of his hands that can beat a bluff.

X, however, must make his ratio of bluffing queens to value betting aces such that Y is indifferent to calling with kings. What we do here is consider the difference in ex-showdown value between Y's two options. When Y calls X's bet with a king, X can hold two different hands. When X holds an ace, X gains one bet when Y calls compared to gaining zero bets if Y folded. When X holds a queen, Y gains not only the bet he called, but the entire pot - since the alternative for Y was to fold and allow X to steal the pot. Since Y wins $(P + 1)$ bets by calling a queen and loses one bet by calling an ace. X must bet one queen for every $P+ 1$ aces.

This can be written as the following equation:

$$b/x = 1/(P+ 1) = \alpha \qquad \textbf{(15.1)}$$

Again, we see that the fundamental a bluffing-to-betting relationship holds. This allows us to ignore the value b in the rest of our analysis, since we can easily find b by simply using the above equation.

Next, let us consider the situation where X checks. Here we have a similar situation to the one in Game #2, except that instead of X having all his aces, he might have bet x of them. Y will bet all his aces and α of his queens, using the same reasoning as we have used throughout. X has some remaining distribution of hands. To make Y indifferent to bluffing, X must fold α of the hands that beat a bluff that remain in his distribution.

If X checked all his aces (suppose $x = 0$), then he would hold equal parts kings and aces. He would then fold α of his remaining hands, selecting kings to fold first. We showed in AXQ Game #2 (Example 13.2) that the fraction of his kings he would fold is $(P-1)/(P+ 1)$. Here, however, his total number of hands has been reduced by x of his aces, so he will need to call with more kings.

The total number of hands that X holds at this point is:

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Total hands = (all his aces) (fraction of aces he checked) + (all kings)
Total hands = $(\frac{1}{2})(1 - x) + (\frac{1}{2})$
Total hands = $\frac{1}{2} - \frac{1}{2}x + \frac{1}{2}$

Fraction to fold = (α) (Total Hands)

$\frac{1}{2}(1 - c) = \alpha(\frac{1}{2} - \frac{1}{2}x + \frac{1}{2})$

$1 - c = \alpha(2 - x)$

$c = 1 - 2\alpha + \alpha x$ **(15.2)**

We can check this formula by looking at two specific values of *x*. First consider *x* = 0%, in which *c* is:

$c = 1 - 2\alpha + \alpha x$
$c = 1 - 2\alpha$
$c = 1 - 2\alpha\,[1/(P+1)]$
$1 - 2\alpha = (P + 1 - 2)/(P + 1)$
$\alpha = (P - 1)/(P + 1)$

as we expected. Alternatively, suppose *x* = 100%. If X bets all his aces, he needs to fold α of his kings to make Y indifferent to bluffing a queen. Our formula shows that $c = 1 - \alpha$. again matching our expectation.

With this equation, then, we know how both sides will play in all situations. However, we still don't know what the value of *x* is.

To recap the solution so far:

X bets *x* of his aces and *αx* of his queens.
X checks and folds the remaining queens and checks and calls with the remaining aces.
X checks and calls with (1 - 2α + αx) of his kings and check-folds the rest.
Y calls with 1 - 2α of his kings and all of his aces if X bets.
Y folds the remaining kings and all queens if X bets.

If X checks, Y bets all his aces and α of his queens, and checks all other hands.

This is a common occurrence when we solve games; often we will obtain optimal solutions except for some particular variable. Then the side that controls that variable will simply select it such that that side's expectation is maximized.

To find *x*, we must find the expectation of the game. We have six cases.

Case 1:
X: A, Y: K.

In this case, X will bet *x* of the time, and Y will call 1 - 2α of the time, for a total value of $-x(1 - 2\alpha)$. When X checks, the value is zero.

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Case 2:

X: A, Y: Q.

In this case, when X bets, the value of the game is zero. X will check and induce a bluff 1 - *x* of the time, and Y will bluff α of the time. The value is thus -α(l - *x).*

Case 3:

X: K, Y: A.

Here X checks all the time. When Y bets, X calls 1 - 2α + *ax* of the time. So the value of this case is 1 - 2α + α*x* to Y.

Case 4:

X:K,Y:Q

Here X checks all the time. Y will bluff a of the time, and when X calls, Y will lose a bet. When X folds, however, Y wins the entire pot. We know that Y's total value from all his bluff must be zero, since X's optimal strategy makes him indifferent between bluffing and checking and the value of checking is zero. So the value of Case 4 plus the value of Case 2 must equal zero. That makes Case 4's value α(l - *x).*

Case 5:

X:Q, Y: A

X bluffs αx of the time. Y also calls for a value of α*x.*

Case 6:

X: Q, Y: K

Again, we know that the total value to the bluffer must be zero from bluffing, since the optimal strategy makes him indifferent to bluffing and checking and the value of checking is zero. So the value of Case 6 must be the negative of Case 5, or -*αx.*

We can view these values as a value matrix:

Hand Matchup	**Value to second player**
(A, K)	-*x* (l – 2α)
(A, Q)	-α (1 – *x*)
(K, A)	1 – 2α + *αx*
(K, Q)	α (1 – *x*)
(Q, A)	αx
(Q, K)	- α*x*
Total	(*x*(-l + 3α) + l - 2α) /6

This last expression gives us the value of the game in terms of x. The following graph shows the

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game value for values of *P* and x:

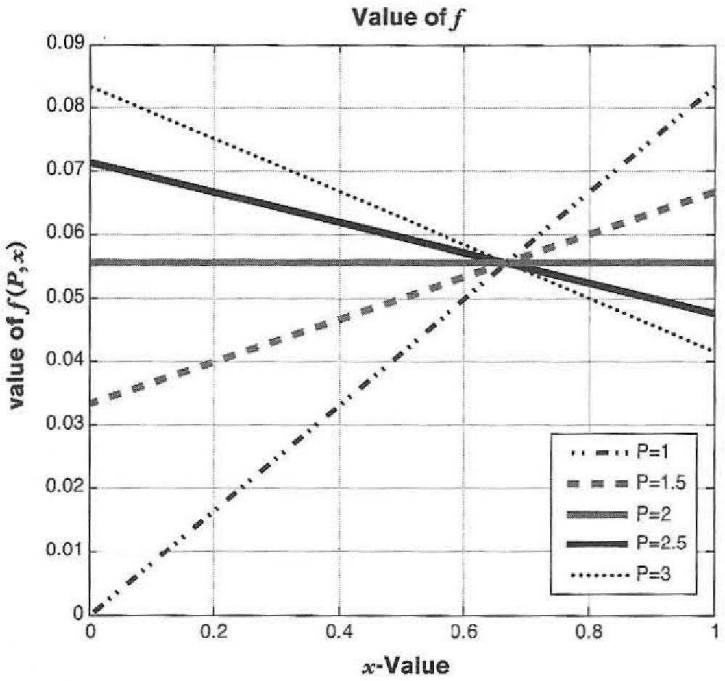

Figure 15.1. Equity of AKQ Game #4 for values of *x.*

The value expression for this game is minimized at x = 1 for values of P larger than 2 and at x = 0 for values of P between 1 and 2.

So we have the following optimal strategies:

Solution:

If P < 2:

Player X:

Checks all hands.
Calls with aces and (*P*- 1)/(*P*+ 1) of his kings.

Player Y:

Bets all aces and 1/(*P*+ 1) of his queens.

If *P > 2:*

Player X:

Bets all aces and 1/*(P*+ 1) of his queens.
Calls with *P/(P*+ 1) of his kings.

Player Y:

If the first player checks, bets all aces and 1/(P+ 1) of his queens.
If the first player bets, calls with aces and *(P- 1)/(P+1)* of hw kings.

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In stating this game, we excluded raising as an option. However, we can verify based on the above solution that if raising is allowed, neither player can improve by raising any of his opponents' bets (except in the cases where he holds an ace and his opponent will fold a queen). Hence this is the solution to the more complete game where raising is allowed as well.

This simple game introduces full-street games. Essentially the differences between these and then half-street counterparts lie in the number of different strategic options available to the players. We saw the recurrence of the same betting to bluffing ratios (α) and the same folding ratio (α again) as we saw in the half-street game. The additional complexity came from X's new decision about whether to induce bluffs or to bet out with his aces.

We turn now back to the no-limit case. Recall that AKQ Game #3 (Example 14.1) was solved in a relatively similar way; we solved the game for strategies in terms of Y's bet size *s* and then maximized Y's equity through his selection of *s*. Having solved the half-street game, it would seem that the jump from the half-street no-limit game to the full-street game should involve only incremental complexity. At the end of this chapter, we proceed to doing just that.

However, before we go on and attempt to solve the AKQ for a full no-limit street, consider the following game.

Example 15.2 - AKQ Game #5

One full street.
Pot size of 4 units.
A special form of spread-limit betting. Either side may check (if no action yet), or bet or raise either 1 or 2 units at their turn.

AKQ Game #5 is the same as AKQ Game #4 (Example 15.1) with pot = 2, except that we've added one additional option for both sides; to bet $^1/_4$ of the pot instead half the pot. When we solved Game #4, we found that at P = 2, X was indifferent to checking or betting aces; hence we know that the value of the game is equal to the value of the half-street game, where X was forced to check dark, since X can check dark in this game without losing value.

In Example 15.1, we found that the value of the game to Y was:

$f(l, P) = (x(-l + 3\alpha) + 1 - 2\alpha)/6$

Here *x* is zero (as X can check dark) and $\alpha = 1/(2 + 1) = {}^1/_3$.

$f(1, 2) = ((0)(-1 + 3\alpha) + 1 - 2({}^1/_3))/6$
$f(l, 2) = (l - {}^2/_3)/6$
$f(1, 2) = {}^1/_{18}$

This result, however, is normalized to a bet size of 1. To correspond it to a bet size of two, we double it:

$f(s = 2, P = 4) = {}^1/_9$

This is the equity of the game if both players simply follow the strategy of playing as though the only allowed bet size is two units. We can now check to see if either player can benefit from betting 1 unit instead. Suppose that X bets 1 unit with his aces and bluffs appropriately with queens. Then Y will respond by making him indifferent to bluffing, which means calling with

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$^4/_5$ of his hands overall, or $^3/_5$ of kings in addition to all his aces. By doing this, X will win 1 unit $^3/_{10}$ of the time, for a total win of $^3/_{10}$ of a unit. Contrast this to his equity when checking. If he checks, Y will bluff $^1/_3$ of his queens (which he holds $^1/_2$ of the time) for a win for the ace of two units, for a total win of $^1/_3$ of a unit. Since $^1/_3$ is larger than $^3/_{10}$, X unilaterally loses equity by betting 1 unit instead of two: hence this is not optimal.

If X checks dark, then Y's equity from changing his bet amount to one unit instead of two:

$f(l, P) = (x(-l + 3\alpha) + 1 - 2\alpha)/6$

So for P = 4:

$f(l, 4) = (l - {}^2/_5)/6$
$f(s = 1, P = 4) = {}^1/_{10}$

Hence Y cannot gain from betting one unit with aces and bluffing appropriately with queens. We also know that Y may raise in this game. However, since X would only bet with aces and queens, neither of which are hands that Y can profitably raise against, Y will just raise with his aces (because calling with them is dominated) and fold queens. The same reasoning applies to X. It seems, then, that this type of game is just a mild extension of the principles we have already studied.

Or is it?

Based on the previous analysis, we have the following strategies:

X:

Check all hands. (This is co-optimal with other strategies.)
Call a 2-unit bet with all aces and $^1/_3$ of kings. Call a 1 unit bet with $^3/_5$ of kings.

Y:

If X bets, call a 2-unit bet with aces and $^1/_3$ of kings and a 1 unit bet with aces and $^3/_5$ of kings.
If X checks, bet two units with all aces and $^1/_3$ of queens.

Recall now the important principle of optimal strategies that ***neither player can improve their expectation unilaterally by changing strategies.*** This means that if we can improve either player's expectation with any hand by playing it differently, the strategies aren't optimal.

Currently, X loses $^2/_3$ of a unit each time he has a king. Half the time, his opponent has a queen and bluffs $^1/_3$ of the time. X calls $^1/_3$ of the time, so $^1/_3$ of the time he gains two units. The other $^2/_3$ of the time, he loses the entire pot of four units, for a total loss when the opponent has a queen of $^2/_3$ of a unit. The other half of the time, the opponent has an ace. In these cases, he loses the bet of two units $^1/_3$ of the time (because he calls). So his total loss when the opponent has an ace is $^2/_3$ of a unit. The average of these, of course, is $^2/_3$.

If we changed his strategy to call more often with kings - for example, to call $^1/_2$ of the time - he would gain value against queens by not losing the pot when the queens bluffed, but lose some value by paying off aces. Since the bluffing to betting ratio for the second player is 3 to 1, the

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total amount he loses with a king remains unchanged.

But what if we bet a king?

In the games we have considered so far, betting a Icing was dominated once we removed the first set of dominated strategies. We would never bet a king because to do so guarantees that we will lose a bet against the opponent's aces while gaining nothing against the opponent's queens. But consider now betting one unit with kings. Our strategy for Y indicates that Y should respond by calling with aces and $^3/_5$ of kings.

If Y does this, then X's total loss with a king is reduced to losing one unit half the time (when he gets called by an ace), for a total loss of $^1/_2$. This loss is smaller than the $^2/_3$ of a unit that X was incurring by checking all his kings. Therefore the strategies presented before cannot be optimal - X can unilaterally improve his equity by betting one unit with kings instead of checking them.

How can Y respond to this bet? We previously excluded raise-bluffing from the strategies, because there were no instances where a player would bet with a king. However, now we have such an occurrence, and so Y must consider raise-bluffing as a non-dominated option. When X bets and Y holds an ace, clearly raising dominates folding. But if he only raises with aces, he cannot gain value because X can simply fold his kings to a raise. So he must also protect his value raises with a corresponding ratio of bluffs.

However, the familiar ratio of *s* to (1 + *s)* no longer applies, because the pot is now bigger (X made it bigger by betting, and the second player must match his bet before raising). There are also two bluff amounts that the second player can choose; a raise of two units (putting in a total of three units) or a raise of one unit (putting in two units total). At first glance, we might think that the ratio of bets to bluffs must make the first player indifferent to calling with a king after betting one unit. However, as we shall see shortly, bluffing that often is more costly for Y. Instead, Y will bluff often enough that X is indifferent to betting one unit with a king and folding to a raise or checking it.

X gained $^2/_3 - ^1/_2 = ^1/_6$of a unit by betting one unit with his lungs instead of checking them. So Y must bluff often enough to make up this difference. Let q be the bluff-raising frequency for the second player (for raising one unit more). Then the second player gains five units each time he bluffs a king successfully. Let q be Y's bluffing frequency.

*(*pot*)p(*Y has a queen) (bluffing frequency) = $^1/_6$
(5)($^1/_2$) $q = ^1/_6$
$q = ^1/_{15}$

If the raise amount is two units, Y bluff-raises $^1/_{15}$ of the time with queens. This makes the first player indifferent to betting one unit or checking kings. The first player should still fold all his kings when the second player raises, because $^{15}/_{16}$of the time, the second player will hold an ace. Notice that it doesn't matter for Y whether he raises one unit or two units when the first player holds a king because the king will fold anyway. However, if X holds an ace, Y would clearly prefer to have raised the smaller amount with a queen.

Now a new strategic option becomes profitable for X. He is indifferent to checking kings or betting them, but his expectation has changed for taking various actions with aces. He had been gaining $^1/_3$ of a unit with aces by checking and picking off Y's bluffs. (Y bluffs $^1/_3$ of his queens, two units, and has a queen $^1/_2$ the time). Now he has a new option, which is to bet one

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unit with aces and an appropriate ratio of queens. This gains him some value from the second player calling with kings. He also gets to pick off some raise-bluffs from the second player.

His expectation from betting one unit with aces is:

<X, bet l > = *p*(Y calls)(bet) + *p(Y* bluff-raises)(raise amount)
<X, bet l > = $(^3/_5)(^1/_2)$(l) + $(^1/_{15})(^1/_2)$(2)
<X, bet l > = $^{11}/_{30}$

His expectation from checking aces is:

<X, check > = *p*(Y bluffs) (bet size)
<X, check > = $(^1/_2)(^1/_3)$(2)
<X, check > = $^1/_3$

So he now gains $^1/_{30}$ of a unit by betting one unit with aces.

Let's review what's happened so far.

We set out to find the optimal strategies for this game with two different bet sizes. Initially, we planned to solve it in the same manner as the previous games, where kings were never bet and the rest was simply maximizing including betting to bluffing ratios. We found that both players ignored the smaller bet and bet aces and queens in ratios as though this were simply Game #4 with a pot of 2. But then a problem arose when we reviewed the strategy. We were able to increase X's equity by having him bet the smaller amount with kings, an option that was previously dominated (when only one bet size was allowed).

We then had to find a value-raising response that would enable Y to avoid being exploited in this way. We found a proper value-raise and corresponding bluffs that made X indifferent to playing this way with kings. But then a new strategic option became profitable for X. He can now bet the smaller amount with his aces and some queens, pick up most of the equity he would have made from checking by getting called by Y's kings, and additionally gain some equity from picking off raise-bluffs.

In all of these strategies, X will only bet queens enough to protect his aces (such that they get called by kings), and so that he will break even on his bluffs with queens.

We know that X has equal equity checking or betting kings (since the second player raise-bluffs enough to ensure this). We gain $^1/_{30}$ of a unit each time by betting aces, so X will want to bet all his aces. We must then find a ratio of kings to aces which forces Y to continue to raise-bluff. Too many kings, and Y profits by raise-bluffing more often; too few, and Y profits by raise-bluffing less often.

The raise bluff gains 5 units when it works (when X has a king), and loses two units when it fails (when X has an ace). So to make Y indifferent to raise-bluffing, X must bet aces and kings in a ratio of 5 to 2. Since X is betting 100% of his aces, he must bet 40% of his kings.

This leads us, then, to a new strategy:

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Solution:

X:

Bet one unit with all aces. Bet one unit with 40% of kings. Bet one unit with 20% of queens.
If Y raises one unit, reraise with aces and fold kings.
If Y raises two units, reraise with aces and fold kings.
With the remaining kings and queens, check.
Call one unit bets $^4/_5$ of the time with kings and two unit bets $^2/_3$ of the time with kings.

Y:

If X bets two units, raise two units with aces and call with $^1/_3$ of kings. Fold queens.
If X bets one unit, raise one unit with aces and $^1/_{15}$ of queens. Call with $^3/_5$ of kings.
If X checks, bet two units with aces and $^1/_3$ of queens. Check kings.

The total value of this game is as follows:

P1 Card	**P2 Card**	**Action**	**Value**	**Probability**	**EV Weight**
A	K	b1 - call	1	3/30	3/30
A	K	bl - fold	0	2/30	0
A	Q	bl - r1 - r -f	2	1/90	1/45
A	Q	bl - fold	0	14/90	0
K	A	bl – rl - f	-1	1/15	-1/15
K	A	chk - b2 - c	-2	1/15	-2/15
K	A	chk - b2 - f	0	1/30	0
K	Q	bl – rl - f	-5	1/225	-1/45
K	Q	bl - f	0	14/225	0
K	Q	chk - b2 - c	2	1/45	2/45
K	Q	chk - b2 - f	4	1/90	-2/45
K	Q	chk - chk	0	1/15	0
Q	A	bl - r2 - f	-1	1/30	-1/30
Q	A	chk - b2 - f	0	2/15	0
Q	K	bl - call	1	1/50	-1/50
Q	K	bl - fold	1	1/75	+1/75
Q	K	chk - chk	0	2/15	0
Total				**1**	**-1/10**

The value of the naive strategy (not involving betting a king) was $^1/_6$. The value of this strategy is - $^1/_{10,}$ a difference of $^1/_{90}$.

It is interesting to note that this is a clear example of information hiding - if X knew his opponent's strategy was fixed, he would prefer to bet two units with his aces instead of one unit. However, with his kings, he prefers to bet he smaller amount. Since he has to choose a single bet size or be exploited, he chooses the smaller amount because it is more valuable (in this case) to make preemptive bets than to bet larger with aces.

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Key Concepts

- It is important not to stop short of the solution when solving games. In this case, we went through the usual process, utilizing results from past games and so on, in order to solve this expanded game. At one point, it seemed we had found what looked like a reasonable solution. Only by going deeper and checking the solutions by comparing the equities of each option did we find an entirely new idea.
- Strategies are only optimal when both sides cannot improve! When we first discovered the above problem (while attempting to solve the full no-limit case), we were quite convinced that we had made a mistake; after all, betting a hand that has no chance of being called by a worse hand and no chance of folding a better hand seems foolish.
- This play with a king is quite out of character with what we see in many other circumstances; bets on the river are "supposed" to be value bets, intended to have positive expectation when they are called, or bluffs, intended to sometimes fold better hands. But this bet is neither. We call it a ***preemptive bet.*** Basically what X is doing is betting and preempting Y's ability to bluff with queens as effectively as he'd like to. Y can respond by raise-bluffing, but now walks into difficulty because X gets more value with his aces. This type of play exists when there are different bet amounts possible and the first player can bet a smaller amount than the second player wants to.
- Betting preemptively is a perfect example of a play that has negative expectation, but the expectation of the play is less negative than the alternative of checking. The important feature is that the value of the strategy is higher than the value of the alternative strategy. It is likely that no-limit holdem on the river contains preemptive betting as part of optimal strategy.
- One other important feature of this game is that after X makes a preemptive bet with a king, he folds all the time to a raise, even though the opponent is bluffing sometimes. This is because he has an excess of strong hands to pick off bluffs with and does not need to call raise bluffs as often. In addition, the raise-bluffer does not attempt to make the original bettor indifferent to calling with a king; instead the raise-bluffer only attempts to make X indifferent to betting a king (vs. checking) in the first place.

With these concepts in mind, we turm our attention now to the full no-limit AKQ game. Because of the complexity and difficulty of this game, we have created an Appendix to Chapter 15, which contains the solution and justification. ***If you understood how the spread-limit solution was derived, then you have understood the most important lesson of the no-limit game.*** At the beginning of the appendix, there is a section that summarizes the steps to find the solution and presents the solution without any justification, The remainder of the appendix contains the mathematics behind the solution, which will likely be of interest primarily to those with a mathematical background.

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